

R6854_Lab2

Loading the Clinical Trial Data

```
library(survival)
data(pbc)
```

Non-trivial transformation/Analysis: finding the mean age for each sex in the trial.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.1      v tibble     3.3.1
v lubridate  1.9.4      v tidyr      1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
pbc |>
  group_by(sex) |>
  summarize(mean_age = mean(age))
```

```
# A tibble: 2 x 2
  sex    mean_age
<fct>   <dbl>
1 m         55.7
2 f         50.2
```

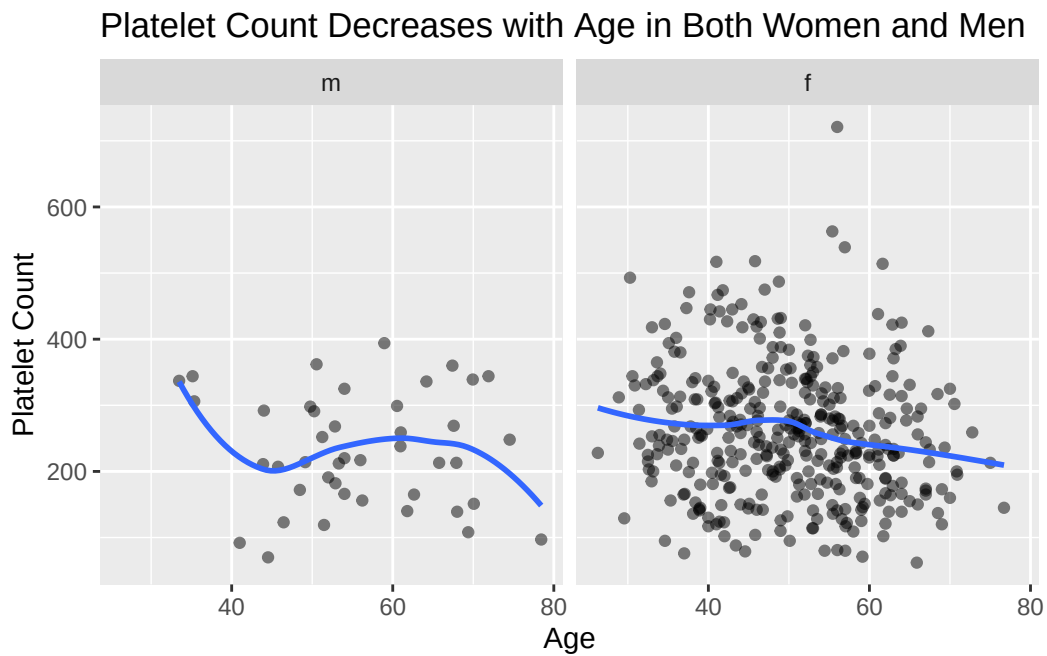
Visualization

```
ggplot(pbc,
       aes(x = age,
           y = platelet)) +
  geom_point(alpha = 0.5) +
  geom_smooth(se = FALSE) +
  facet_wrap(~sex) +
  labs(title = "Platelet Count Decreases with Age in Both Women and Men", x="Age",y="Platelet Count")
```

``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

Warning: Removed 11 rows containing non-finite outside the scale range (``stat_smooth()``).

Warning: Removed 11 rows containing missing values or values outside the scale range (``geom_point()``).



Interpretation

The plot shows that platelet count generally decreases with age. Platelet count seems to decrease more consistently among females as they age, but overall seems to decrease less over time than their male counterparts. This visualization also shows that there are more females in the clinical trial than men.

Comparison between R and Python

I find python code easier to write usually, but I find it easier to create visuals in R. I also trust the visual output of analyses from R more, as it is catered more for statistical analyses. I was able to produce the code for the visualization in R much more quickly because all visualizations in R follow the same structure in ggplot2. I had to use AI to find code to create the same visualiztion in Python.